

# **DS605 –Lab Assignment – 4**

## **End-to-End Machine Learning Project: Airbnb Price Prediction**

### **Task 4 – Final Project Summary**

| <b>Project Detail</b> | <b>Information</b>        |
|-----------------------|---------------------------|
| Student Name          | Ameya Trivedi             |
| Enrollment Number     | 202618047                 |
| Course                | DS605 – Machine Learning  |
| Dataset               | AB_NYC_2019.csv           |
| Dataset Size          | 48,895 rows × 16 columns  |
| Application           | Streamlit                 |
| Deployment            | Streamlit Community Cloud |

## Project Objective

The objective of this project was to complete an end-to-end machine learning workflow for Airbnb price prediction. The project covers data analysis and preparation, model training and evaluation, and development of a deployed Streamlit application that accepts Airbnb listing information and returns an estimated nightly price.

**Live Application Link:** [Live Application Link](#)

## TASK 1 – DATA ANALYSIS AND PREPARATION

### 1.1 Dataset and Initial Analysis

The Airbnb dataset used for this project is AB\_NYC\_2019.csv, containing 48,895 New York City Airbnb listings and 16 original columns. The variables include listing and host information, neighbourhood and location, room type, price, minimum nights, reviews, host listing count, and availability.

### 1.2 Data Cleaning

The dataset was inspected for missing values, duplicate records, invalid target values, and unusual observations. Initially, missing values were present in last\_review, reviews\_per\_month, host\_name, and name. The review-related missingness corresponded to listings with no reviews. Missing values in the required fields were handled, last\_review was dropped because it was not used by the final model, and duplicate records were checked.

| Cleaning Check                                    | Result     |
|---------------------------------------------------|------------|
| Original rows                                     | 48,895     |
| Original columns                                  | 16         |
| Missing values                                    | Handled    |
| Duplicate records                                 | None       |
| Zero-price listings                               | 11 removed |
| Final modelling columns after last_review removal | 15         |

### 1.3 Missing Values

| Column            | Missing Values | Percentage |
|-------------------|----------------|------------|
| last_review       | 10,052         | 20.56%     |
| reviews_per_month | 10,052         | 20.56%     |
| host_name         | 21             | 0.04%      |
| name              | 16             | 0.03%      |

### 1.4 Outlier Analysis

The price variable was strongly right-skewed. After removing the 11 listings with price equal to zero, the cleaned price distribution had a median of \$106, a mean of approximately \$152.76, and a maximum of \$10,000.

| Statistic                         | Price    |
|-----------------------------------|----------|
| Q1                                | \$69     |
| Median                            | \$106    |
| Q3                                | \$175    |
| IQR                               | \$106    |
| Upper IQR boundary                | \$334    |
| Observations above upper boundary | 2,972    |
| Maximum                           | \$10,000 |

High-price observations were investigated rather than being automatically deleted. Some high-priced listings can represent legitimate premium or luxury properties. Therefore, the final workflow used log transformation of the target to reduce skewness and the influence of extreme values.

### 1.5 Exploratory Data Analysis and Patterns

Exploratory analysis included price distributions, price by room type, price by neighbourhood group, and relationships between price and numerical variables.

| Group           | Listings | Mean Price | Median Price |
|-----------------|----------|------------|--------------|
| Entire home/apt | 25,407   | \$211.81   | \$160        |
| Private room    | 22,319   | \$89.81    | \$70         |
| Shared room     | 1,158    | \$70.25    | \$45         |
| Manhattan       | 21,660   | \$196.88   | \$150        |
| Brooklyn        | 20,095   | \$124.44   | \$90         |
| Queens          | 5,666    | \$99.52    | \$75         |
| Staten Island   | 373      | \$114.81   | \$75         |
| Bronx           | 1,090    | \$87.58    | \$65         |

The analysis indicates that room type and location are important price-related patterns. Entire homes/apartments had the highest average price among room types, while Manhattan had the highest average price among neighbourhood groups.

### 1.6 Feature Engineering

To reduce skewness in numerical variables, log1p transformations were created for minimum\_nights, number\_of\_reviews, reviews\_per\_month, calculated\_host\_listings\_count, and availability\_365.

log\_minimum\_nights

log\_number\_of\_reviews

log\_reviews\_per\_month

log\_host\_listings\_count

log\_availability\_365

### 1.7 Feature Selection

The final 10 features selected for model training were:

neighbourhood\_group

neighbourhood

latitude

longitude

room\_type

log\_minimum\_nights

log\_number\_of\_reviews

log\_reviews\_per\_month

log\_host\_listings\_count

log\_availability\_365

The target variable price was transformed using log1p(price) for the final model training workflow. The data was split into training and test sets using a 80:20 split, producing 39,107 training observations and 9,777 test observations.

## TASK 2 – MODEL TRAINING AND EVALUATION

### 2.1 Regression Models Trained

Three suitable regression approaches were trained and compared: Linear Regression, Random Forest, and Gradient Boosting. Models were evaluated using MAE, RMSE, and R<sup>2</sup>.

| Model                               | MAE   | RMSE   | R <sup>2</sup> |
|-------------------------------------|-------|--------|----------------|
| Linear Regression – Original Target | 70.97 | 185.02 | 0.145          |
| Random Forest – Original Target     | 66.87 | 196.74 | 0.033          |
| Gradient Boosting – Original Target | 66.07 | 181.45 | 0.178          |

## 2.2 Log-Target Model Comparison

| Model                          | MAE   | RMSE   | R <sup>2</sup> |
|--------------------------------|-------|--------|----------------|
| Random Forest – Log Target     | 56.32 | 175.14 | 0.234          |
| Gradient Boosting – Log Target | 57.49 | 184.30 | 0.152          |
| Linear Regression – Log Target | 59.20 | 185.84 | 0.137          |

The log-target Random Forest produced the best overall test performance among the evaluated models. It achieved the lowest test MAE and RMSE and the highest test R<sup>2</sup> within the log-target comparison. This model was selected as the final model.

## 2.3 Hyperparameter Tuning

GridSearchCV was used for hyperparameter tuning of the Random Forest model. A 3-fold cross-validation procedure evaluated 15 parameter combinations, resulting in 45 fits.

| Hyperparameter    | Best Value |
|-------------------|------------|
| n_estimators      | 200        |
| min_samples_split | 2          |
| min_samples_leaf  | 2          |
| max_features      | 0.7        |
| max_depth         | 20         |

Best cross-validation RMSE on the log scale: 0.4339.

## 2.4 Tuned Model Evaluation

| Metric         | Tuned Model Test Result |
|----------------|-------------------------|
| MAE            | 55.32                   |
| RMSE           | 176.97                  |
| R <sup>2</sup> | 0.218                   |

Although tuning reduced the test MAE slightly, the tuned model had a higher RMSE and lower R<sup>2</sup> than the baseline log-target Random Forest. Therefore, the baseline log-target Random Forest was retained as the final model because it had better overall test performance.

## 2.5 Final Model and Overfitting Check

| Metric         | Training | Test   |
|----------------|----------|--------|
| MAE            | 25.57    | 56.32  |
| RMSE           | 157.13   | 175.14 |
| R <sup>2</sup> | 0.602    | 0.234  |

The difference between training and test performance shows that the model fits the training data more strongly than the unseen test data. The test R<sup>2</sup> of 0.234 indicates that the selected features explain a meaningful but limited portion of price variation. This was considered when interpreting the final model.

## 2.6 Saving the Final Preprocessing Workflow and Model

The final preprocessing and trained model workflow was saved using joblib so that the same transformations and trained estimator could be reused by the Streamlit application. The deployed project uses the compressed saved model file:

**airbnb\_price\_model\_compressed.pkl**

## TASK 3 – STREAMLIT OR GRADIO APPLICATION

### 3.1 Application Development

A Streamlit web application was developed to provide a simple interface for Airbnb price prediction. The application accepts Airbnb listing information and returns the estimated nightly price generated by the final trained model.

### 3.2 User Inputs

Neighbourhood Group

Neighbourhood

Room Type

Latitude

Longitude

Minimum Nights

Number of Reviews

Reviews per Month

Host Listings Count

Availability (Days Per Year)

### 3.3 Prediction Workflow

When the user selects or enters listing information, the application creates the same engineered features used during training. The numerical variables are log-transformed using log1p, the saved

preprocessing/model workflow generates a prediction on the log-price scale, and expm1 is used to convert the prediction back to the original nightly price scale.

### 3.4 Application Testing with Realistic Input

The deployed application was tested using a realistic Manhattan Airbnb listing:

| Input               | Value           |
|---------------------|-----------------|
| Neighbourhood Group | Manhattan       |
| Neighbourhood       | Midtown         |
| Room Type           | Entire home/apt |
| Latitude            | 40.75           |
| Longitude           | -73.98          |
| Minimum Nights      | 3               |
| Number of Reviews   | 10              |
| Reviews per Month   | 1.00            |
| Host Listings Count | 1               |
| Availability        | 100             |

**Application Result:** Estimated Airbnb Price = \$ 260.63 per night.

### 3.5 Online Deployment

The Streamlit application was successfully deployed online using Streamlit Community Cloud and tested through the deployed interface.

**Live Application Link:** [Live Application Link](#)

## TASK 4 – FINAL PROJECT SUMMARY

### 4.1 Analysis Summary

The Airbnb dataset was cleaned, explored, and prepared for machine learning. Missing values and invalid zero-price observations were handled, duplicate records were checked, outliers were investigated, and skewed numerical variables were transformed. Exploratory analysis showed clear differences in prices by room type and neighbourhood group.

### 4.2 Model Comparison Summary

Linear Regression, Random Forest, and Gradient Boosting were compared. Using a log-transformed target improved the Random Forest result. The final selected model was the Random Forest with log-transformed price because it provided the best overall test performance among the evaluated models.

### 4.3 Final Performance

The final model achieved a test MAE of 56.32, RMSE of 175.14, and  $R^2$  of 0.234. The model was saved and integrated into the deployed application.

#### 4.4 Application Results

The Streamlit application successfully accepted realistic Airbnb listing information and returned a nightly price estimate. For the realistic Manhattan/Midtown test case, the application predicted \$226.06 per night.

#### 4.5 Limitations

The test  $R^2$  of approximately 0.234 indicates that substantial price variation remains unexplained.

The dataset represents New York City Airbnb listings from 2019 and may not represent current market prices.

The final model does not include every possible price-influencing factor, such as detailed amenities, seasonality, property quality, or current demand.

Legitimate premium listings make simple removal of all high-price outliers inappropriate.

The application provides an estimate and should not be treated as an exact market price.

#### 4.6 Conclusion

The project completed the required end-to-end machine learning workflow for Airbnb price prediction. Data preparation and exploratory analysis were followed by feature engineering, model comparison, hyperparameter tuning, final model selection, and deployment. The final Random Forest model with a log-transformed target was integrated into a Streamlit application and successfully tested with realistic input. The project demonstrates how a machine learning model can be taken from raw data through evaluation to a usable online prediction application.

#### Project Deliverables

| Deliverable                            | Status             |
|----------------------------------------|--------------------|
| Task 1 – Data Analysis and Preparation | Completed          |
| Task 2 – Model Training and Evaluation | Completed          |
| Task 3 – Streamlit/Gradio Application  | Completed          |
| Task 3 – Online Deployment             | Completed          |
| Task 4 – Final Project Summary         | Completed          |
| GitHub Repository                      | Updated and pushed |